

PART 1:

Dataset Source

I am using the same dataset as Project 1.

The dataset was downloaded from Kaggle:

<https://www.kaggle.com/datasets/raminhuseyn/demand-forecasting-dataset>

It is a demand forecasting dataset that simulates retail sales data across multiple stores, products, and regions.

Data Description

This dataset represents retail business operations and includes information about product demand, sales, pricing, and external factors.

It allows analysis of:

- Demand vs actual sales performance
- Differences across products and regions
- Impact of pricing and external factors

The dataset includes both transactional data and supporting variables such as category, region, and pricing, which help explain variations in demand.

Categorical Variables

- Store ID – Identifies different store locations
- Product ID – Represents individual products
- Category – Groups products into categories
- Region – Indicates geographical location

- Weather Condition – External environmental factor affecting demand

Numerical Variables

- Demand – Number of units required
- Units Sold – Number of units sold
- Units Ordered – Quantity of products ordered
- Price – Cost per unit
- Discount – Reduction applied to price
- Inventory Level – Available stock
- Competitor Pricing – Price offered by competitors

Temporal Variable

- Date – Represents the time of transaction and is used for trend analysis

Variable Summary

The dataset includes three types of variables:

- Categorical variables for grouping and comparison
- Numerical variables for calculations and analysis
- Temporal variables for time-based analysis

This combination allows analysis across different dimensions such as time, location, and product characteristics.

Work Completed in Project 1

In Project 1, I used this dataset to build a dashboard focused on analyzing demand and sales performance.

The main work included:

- Identifying and classifying variables into categorical, numerical, and temporal types
- Creating key DAX measures such as Total Demand, Total Sales, Average Price, and Total Orders
- Building visualizations to analyze trends, comparisons, and distributions

The dashboard included:

- A line chart to analyze demand trends over time
- Column charts to compare store and product performance
- A pie chart to examine regional demand distribution
- A table to analyze product-level details such as demand, sales, and pricing

From this analysis, I identified key insights such as:

- Demand fluctuations over time
- Balanced performance across stores
- Demand distribution across products
- Regional differences in demand
- Limited direct relationship between price and demand

These insights helped build a strong understanding of the dataset, which was further expanded in Project 2 using more advanced visualizations and dashboard design techniques.

PART 2:

KPI CARDS

KPI Selection

For this project, I chose to enhance and extend the KPIs from Project 1 rather than keep them in their basic form.

In Project 1, the KPIs focused mainly on simple aggregates such as Total Demand and Total Sales. While useful, they only provided a summary view and did not explain performance, efficiency, or business impact.

In Project 2, the goal is to build a more analytical dashboard. Therefore, I introduced additional KPIs that measure:

- Performance against expectations
- Operational efficiency
- Business drivers such as promotions

This allows the dashboard to move from descriptive analysis (what happened) to analytical insight (why it happened).

KPIs Used in This Dashboard

The dashboard presents key performance metrics using four KPI cards, each combining a primary metric with supporting values:

- Total Units Sold (with supporting metrics: Total Demand and Demand Gap)
- Fill Rate % (with supporting metrics: Total Units Sold and Total Demand)
- Promotion % (with supporting metrics: Promotion Sales and Total Units Sold)
- Total Sales (with supporting metrics: Total Units Sold and Average Price)

DAX Formulas Used

Total Units Sold = SUM(demand_forecasting[Units Sold])

Total Demand = SUM(demand_forecasting[Demand])

Demand Gap = [Total Units Sold] - [Total Demand]

Fill Rate % = DIVIDE([Total Units Sold], [Total Demand])

Promotion Sales = CALCULATE(
 [Total Units Sold],
 demand_forecasting[Promotion] = 1
)

Promotion % = DIVIDE([Promotion Sales], [Total Units Sold])

Avg Price = AVERAGE(demand_forecasting[Price])

Explanation of KPI Logic

- Units Sold vs Demand compares actual sales (Total Units Sold) with expected demand (Total Demand), while Demand Gap highlights the shortfall between them
- Fill Rate % measures how effectively demand is being fulfilled using Total Units Sold and Total Demand

- Promotion % shows the proportion of sales driven by promotions, supported by Promotion Sales and Total Units Sold
- Total Sales provides an overall summary of sales performance, supported by Units Sold and Average Price for additional context

These metrics are grouped into KPI cards to provide both high-level performance indicators and supporting context, allowing users to quickly understand overall performance, efficiency, and key business drivers.

Advanced KPI Features (Chapter 18)

1. Multi-Measure KPI Cards

Each KPI card combines a primary metric with supporting values.

For example:

- Units Sold is shown together with Total Demand and Demand Gap
- Fill Rate % is supported by Units Sold and Total Demand
- Promotion % is supported by Promotion Sales and Units Sold
- Total Sales is supported by Units Sold and Average Price

This approach provides more context within a single visual and reduces the need for additional charts.

2. Conditional Formatting

Conditional formatting is used to improve interpretation:

- Negative Demand Gap indicates underperformance

- Percentage-based KPIs such as Fill Rate % and Promotion % help quickly evaluate performance levels

This allows users to identify issues at a glance without detailed analysis.

3. Comparison Indicators

The KPI cards are designed to compare actual performance with expected values:

- Units Sold vs Demand shows performance against expectations
- Fill Rate % acts as a performance ratio
- Demand Gap highlights deviation from expected demand

This helps users understand whether business performance is meeting targets.

4. Professional Styling

The KPI cards follow a consistent and clean design:

- White background with light borders
- Consistent font and color usage
- Clear hierarchy between main values and supporting values

This improves readability and ensures a professional dashboard appearance.

Why These KPIs Were Used

These KPI cards were selected to provide a structured overview of performance from multiple perspectives:

- Performance → Units Sold vs Demand
- Efficiency → Fill Rate %

- Business driver → Promotion %
- Sales & pricing context → Total Sales with Average Price

By combining primary metrics with supporting values, the dashboard allows users to quickly understand both overall performance and the key factors influencing it.

Role of KPIs in the Dashboard

The KPI cards represent Level 1 of the dashboard.

They provide a quick summary of key metrics, allowing users to:

- Understand overall performance immediately
- Identify issues such as unmet demand (Demand Gap)
- Evaluate efficiency using Fill Rate %
- Understand the impact of promotions and pricing

This structure follows a top-down analytical approach, where users begin with high-level insights and then move to detailed visualizations for deeper analysis.

PART 3:

VISUALIZATION JUSTIFICATION

Classic Visualizations

1. Column Chart (Price Analysis)

- Chapter: Chapter 13 (Comparison Visuals)
- Visualization Principle (Chapter 2): Ranking

Explanation:

This visualization shows the ranking of product categories based on average price. It allows easy comparison across categories and highlights which categories are priced higher or lower.

This type of chart makes sense for the data because price is a numerical variable that needs to be compared across categories. A column chart provides a clear and intuitive way to display differences and identify patterns.

Advanced Visualizations

1. KPI Cards

- Chapter: Chapter 18

Why it works:

KPI cards provide a high-level summary of key business metrics such as Units Sold, Demand, Fill Rate %, and Promotion %. They allow users to quickly understand overall performance without analyzing detailed charts.

Data preparation:

DAX measures were created to calculate aggregated values and performance metrics.

DAX formulas used:

Total Units Sold = SUM(demand_forecasting[Units Sold])

Total Demand = SUM(demand_forecasting[Demand])

Demand Gap = [Total Units Sold] - [Total Demand]

Fill Rate % = DIVIDE([Total Units Sold], [Total Demand])

Promotion Sales = CALCULATE(

```
[Total Units Sold],  
demand_forecasting[Promotion] = 1  
)
```

Promotion % = DIVIDE([Promotion Sales], [Total Units Sold])

Avg Price = AVERAGE(demand_forecasting[Price])

2. Bullet Chart

- Chapter: Chapter 15
- Visualization Principle (Chapter 2): Overview

Why it works:

The bullet chart compares actual performance (Units Sold) against a target (Demand). It provides a compact and effective way to evaluate whether performance meets expectations across categories.

This visualization is appropriate because the dataset includes both actual and target values, making it ideal for performance comparison.

Data preparation:

Used existing measures:

- Total Units Sold
- Total Demand

No additional transformation was required.

3. Waterfall Chart

- Chapter: Chapter 14
- Visualization Principle (Chapter 2): Structure

Why it works:

The waterfall chart shows how promotion-driven sales contribute to total sales across categories.

It helps break down overall performance into meaningful components.

This visualization is suitable because it highlights how different categories contribute to total promotional sales, providing insight into business drivers.

Data preparation:

Used Promotion Sales measure.

DAX used:

```
Promotion Sales = CALCULATE(  
    [Total Units Sold],  
    demand_forecasting[Promotion] = 1  
)
```

4. Sankey Diagram

- Chapter: Chapter 17
- Visualization Principle (Chapter 2): Structure

Why it works:

The Sankey diagram visualizes the flow of sales from regions to categories. It helps identify how different regions contribute to category-level sales.

This is useful because the dataset includes categorical relationships (Region → Category), and Sankey effectively represents these flows.

Data preparation:

Fields used:

- Source → Region
- Destination → Category
- Weight → Total Units Sold

No additional DAX was required.

Summary

The selected visualizations were chosen to provide a balanced dashboard that covers multiple analytical perspectives:

- Ranking → Column chart (price comparison)
- Overview → KPI cards and bullet chart
- Structure → Waterfall and Sankey

Each visualization was selected based on the nature of the data and the type of insight required, ensuring clarity, non-redundancy, and effective communication.

PART 4:

DASHBOARD DESIGN AND LAYOUT

Level 0 – Header Area

The dashboard includes a header section at the top consisting of a title and a date slicer.

The title “Demand Forecasting Dashboard” is placed on the top-left to clearly communicate the purpose of the dashboard. A date slicer is placed on the top-right, allowing users to filter the entire dashboard based on a selected time range.

This dashboard is designed as a single-page layout, so navigation elements are not required. A company logo was not included to maintain a clean and focused design.

Level 1 – KPI Cards

The first level of the dashboard contains key performance indicators displayed as KPI cards.

The KPIs are grouped into four main cards:

- Units Sold (with supporting metrics: Total Demand and Demand Gap)
- Fill Rate % (with supporting metrics: Units Sold and Total Demand)
- Promotion % (with supporting metrics: Promotion Sales and Units Sold)
- Total Sales (with supporting metrics: Units Sold and Average Price)

These KPIs were selected to provide a quick overview of performance, efficiency, and business impact.

All KPI cards are:

- Styled consistently using a white background and light borders
- Aligned horizontally for easy comparison
- Designed with clear visual hierarchy, where primary values are larger than supporting values

Level 2 – Summary Visualizations

The second level contains summary-level visualizations:

- Bullet Chart (Units Sold vs Demand by Category)
- Waterfall Chart (Promotion Sales Contribution by Category)

The bullet chart compares actual performance against expected demand across categories, allowing quick evaluation of performance.

The waterfall chart shows how each category contributes to total promotion sales, helping identify key drivers.

Both visuals support cross-filtering, allowing users to interact with categories and update other visuals dynamically.

Level 3 – Detailed Information

The third level provides detailed analytical insights:

- Column Chart (Price Variation Across Categories)
- Sankey Diagram (Sales Flow from Region to Category)

The price chart shows differences in average price across categories, helping identify pricing patterns.

The Sankey diagram visualizes the flow of sales from regions to categories, showing relationships and distribution of sales.

This level allows users to explore detailed patterns and relationships not visible in summary views.

Grid Layout

A 2×2 grid layout is used in this dashboard.

- Top row → KPI cards

- Middle row → Summary visuals
- Bottom row → Detailed visuals

This layout was chosen because:

- It provides a clear top-down analytical flow
- It separates summary insights from detailed analysis
- It ensures proper alignment and readability

Theme and Styling

A consistent and professional theme is applied across the dashboard.

Color Palette:

- Blue → primary data color
- Green → positive contributions (waterfall)
- Light gray → background and borders
- White → visual containers

Fonts:

- Consistent font style across all visuals
- Larger font for titles and KPI values
- Smaller font for supporting information

Background:

- Light gray page background
- White visual containers with subtle borders

This creates a clean and modern appearance while maintaining readability.

Cross-Filtering Setup

Cross-filtering is enabled across all visuals.

- Selecting a category or region updates all other visuals
- KPI cards, summary visuals, and detailed visuals respond dynamically

Filtering is used instead of highlighting to maintain clarity and avoid visual clutter.

This allows users to interactively explore relationships between performance, pricing, promotion impact, and regional distribution.

These are the screen shots for the information discussed above :

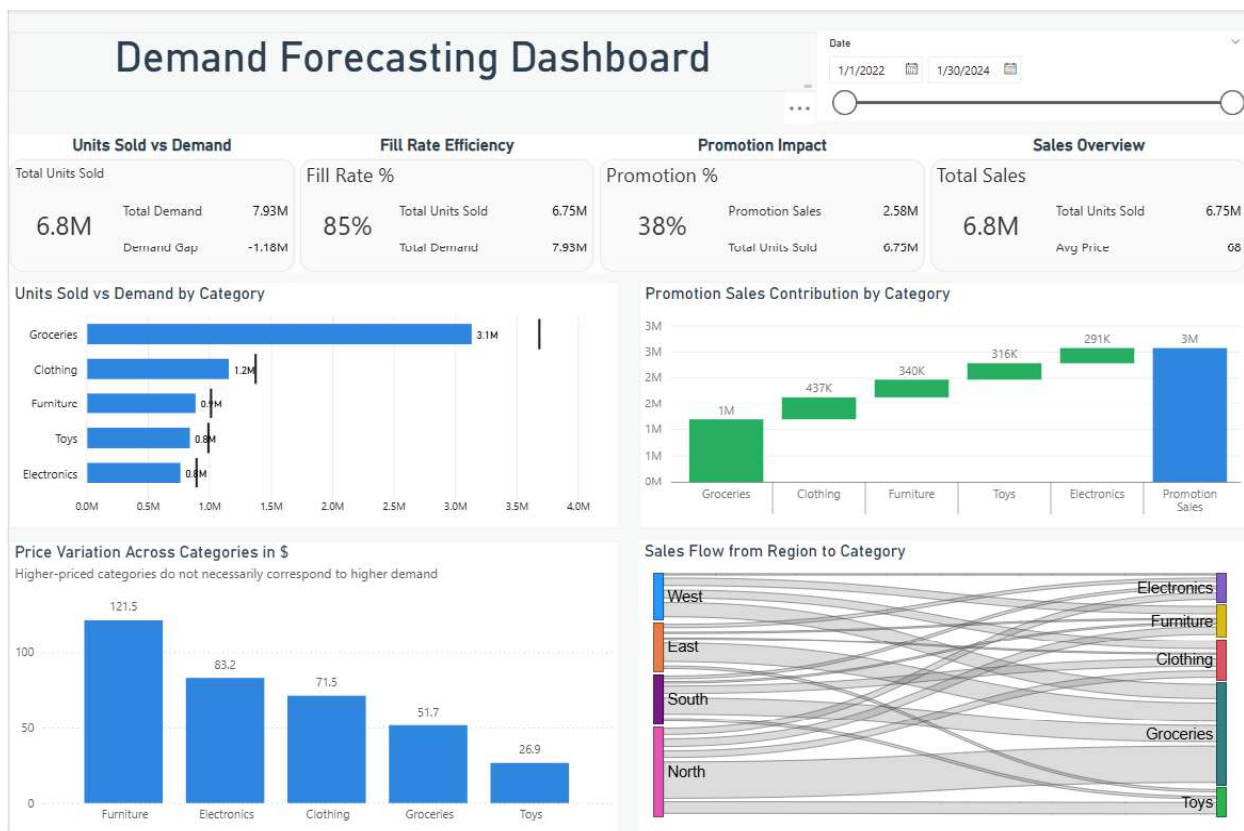


Figure 1: Full dashboard showing the 2x2 grid layout

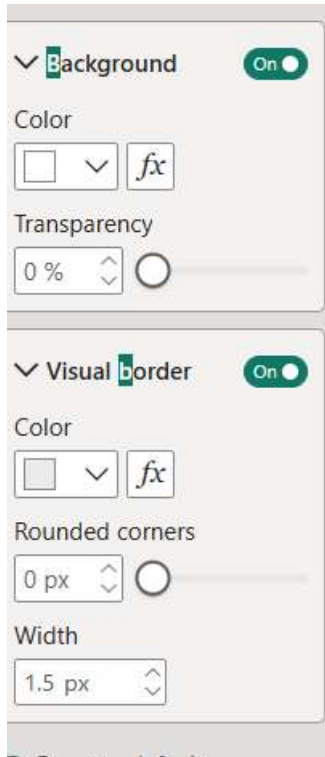


Figure 2: Theme and color settings (same across all the visualizations)

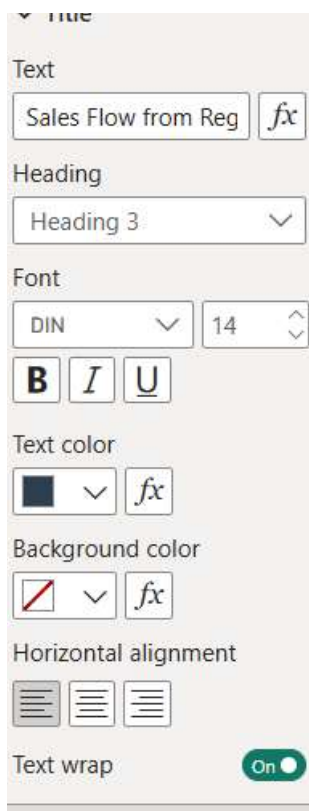
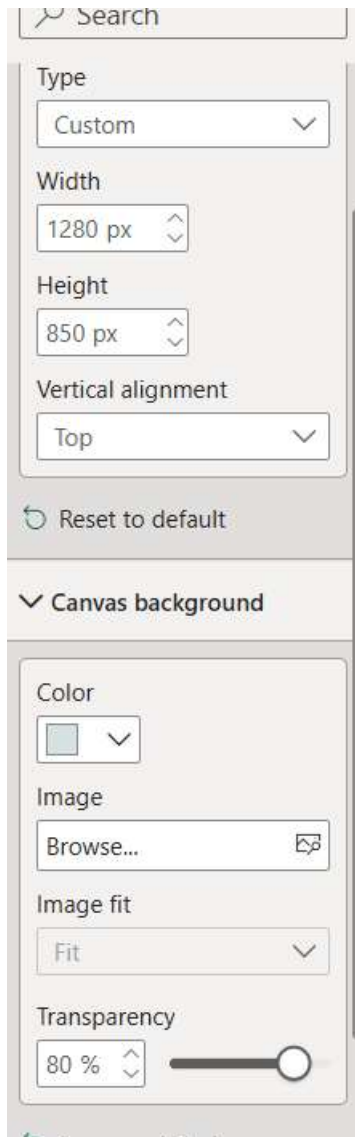


Figure 3: Title and color settings (same across all the visualizations)

Figure



4:Background and visual styling

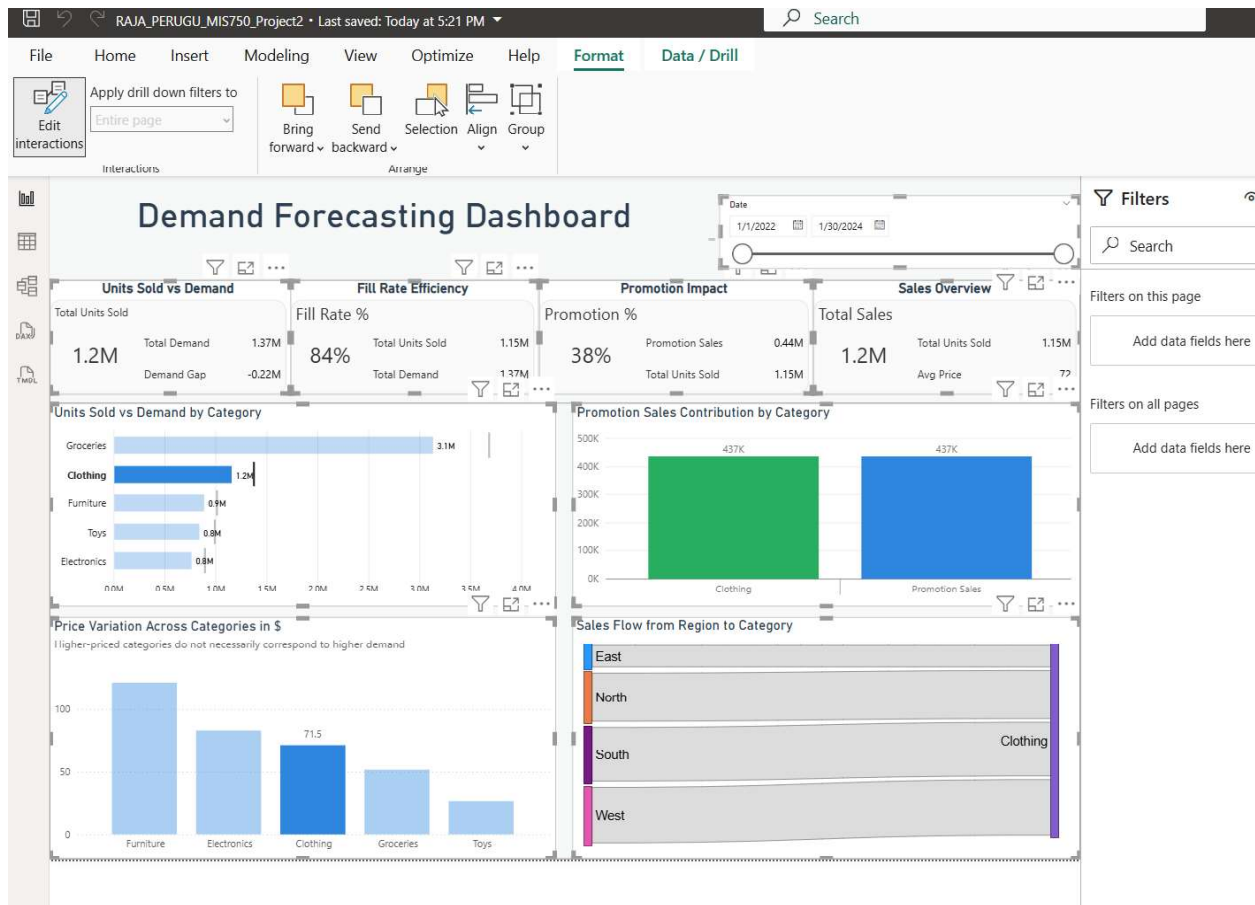


Figure 5: Cross-filtering setup using "Edit Interactions" mode

PART 5: DASHBOARD PITCH

This dashboard is designed for business managers, operations teams, and analysts who are responsible for monitoring sales performance, demand forecasting, and inventory planning. It is especially useful for decision-makers who need a quick overview of how well demand is being met and what factors are driving performance across categories and regions.

The main business problem this dashboard addresses is the gap between forecasted demand and actual sales, along with understanding the role of pricing and promotions in influencing

performance. By highlighting metrics such as fill rate, demand gap, and promotion impact, the dashboard helps identify inefficiencies, missed opportunities, and areas where demand is not being fully satisfied.

Users can interact with the dashboard using the date filter to analyze performance over different time periods. They can also click on categories or regions within the visuals to dynamically update the entire dashboard through cross-filtering. This allows users to explore specific segments, compare performance across categories, and understand relationships between variables such as pricing, promotions, and regional sales distribution.

This dashboard is valuable to an organization because it provides a clear, structured view of performance from high-level KPIs down to detailed insights. It supports faster decision-making by identifying where improvements are needed, such as increasing inventory in high-demand regions or optimizing promotional strategies. Overall, it helps organizations make more informed, data-driven decisions to improve efficiency and sales performance.

Insight 1: Overall Sales Performance (KPI Summary)

Insight Statement 1

The business sold 6.8M units against a total demand of 7.93M units, resulting in a demand gap of -1.18M units and a fill rate of 85%. Additionally, 38% of total sales are driven by promotions, indicating a strong reliance on promotional activities.

Screenshot

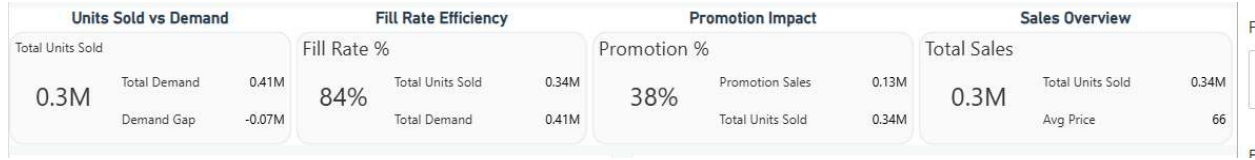


Figure 6:KPI's used in Dashboard

Business Value

This insight shows that the company is not fully meeting customer demand, which leads to missed revenue opportunities. The high dependence on promotions suggests that a significant portion of sales may not be sustainable without discounts, potentially affecting profit margins.

Actionable Recommendation

- Improve demand fulfillment by increasing inventory or optimizing supply chain operations
- Reduce reliance on promotions by strengthening pricing strategy or product positioning
- Monitor fill rate regularly to ensure demand is being met efficiently

Insight 2 : Units Sold vs Demand by Category

Insight Statement

Groceries significantly outperform all other categories in terms of units sold, but still show a gap compared to demand. Clothing shows moderate performance, while Furniture, Toys, and Electronics have relatively similar and lower sales levels. Across all categories, actual units sold are consistently below demand, indicating unmet demand.

Screenshot

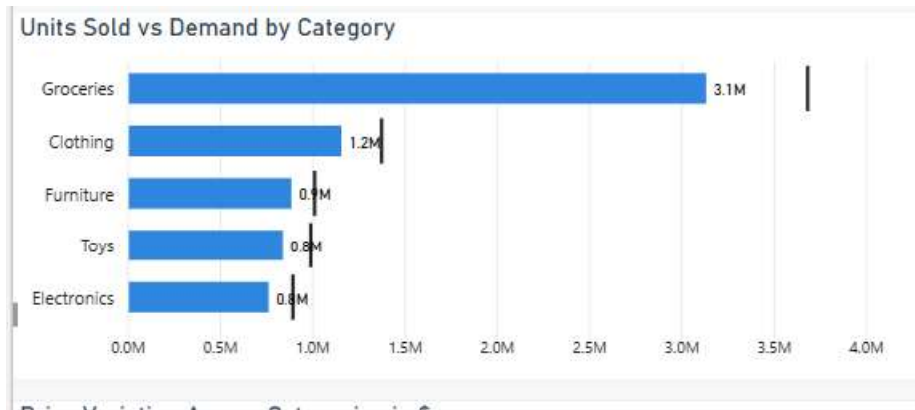


Figure 7:Bullet Chart

Business Value

This insight highlights that the business is not fully meeting customer demand across all categories, especially in high-demand categories like Groceries. This can lead to lost revenue opportunities and reduced customer satisfaction. It also shows which categories contribute most to overall sales performance.

Actionable Recommendation

- Increase inventory and supply for high-demand categories like Groceries to reduce the demand gap
- Investigate supply chain or stock limitations causing unmet demand
- Evaluate whether pricing or promotions can help improve sales in lower-performing categories like Electronics and Toys

Insight 3: Promotion Sales Contribution by Category

Insight Statement

Groceries contribute the largest portion of promotion-driven sales, followed by Clothing, Furniture, Toys, and Electronics. Promotion sales are built cumulatively across categories, reaching a total of approximately 3M units, with Groceries alone contributing a significant share.

Screenshot

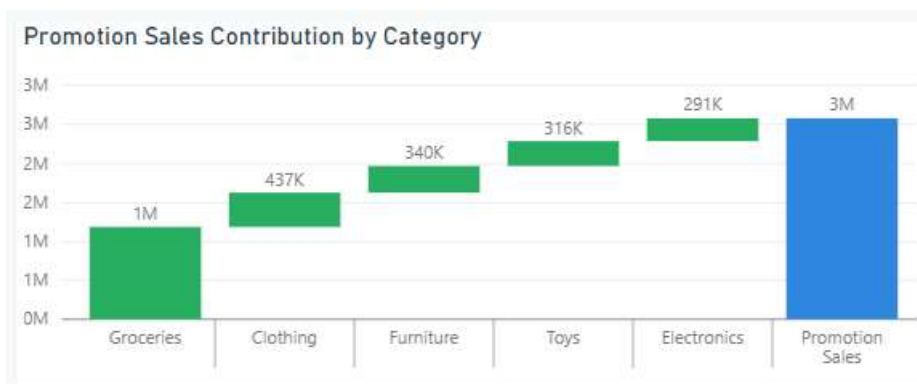


Figure 8: Waterfall Chart

Business Value

This insight shows that promotions are highly effective in certain categories, especially Groceries. It highlights which categories respond most to promotional strategies and contribute the most to overall promotional sales. This helps identify where marketing efforts are generating the highest returns.

Actionable Recommendation

- Focus promotional campaigns on high-performing categories like Groceries and Clothing to maximize impact

- Analyze why categories like Electronics and Toys contribute less and adjust promotion strategies accordingly
- Optimize promotional spending by prioritizing categories with the highest return on investment

Insight 4: Price Variation Across Categories

Insight Statement:

Furniture has the highest average price (around \$121), followed by Electronics (\$83) and Clothing (\$71), while Groceries (\$52) and Toys (\$27) are significantly lower. Despite these differences, higher-priced categories do not necessarily correspond to higher sales or demand.

Screenshot

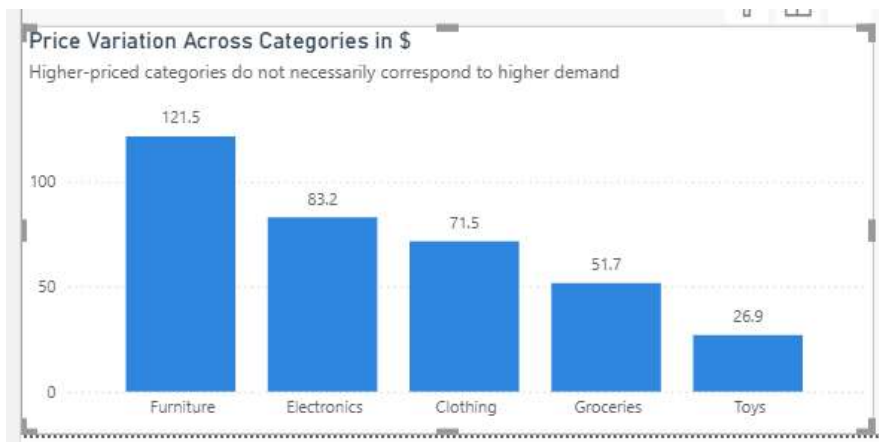


Figure 9: Price Variation Chart

Business Value

This insight shows that price alone does not drive demand. Customers are still purchasing mid- and lower-priced categories in higher volumes, while higher-priced categories like Furniture do not dominate sales. This indicates that other factors such as product type, necessity, and promotions play a more important role in influencing demand.

Actionable Recommendation

- Avoid relying solely on price reductions to increase demand
- Use targeted promotions or bundling strategies for higher-priced categories like Furniture and Electronics
- Focus on maintaining competitive pricing for mid-range categories like Clothing, which balance price and demand effectively

Insight 5: Sales Flow from Region to Category

Insight Statement:

The North region contributes the largest share of sales across most categories, particularly in Groceries, while other regions such as East, South, and West show more balanced but comparatively lower contributions. Groceries receive the highest overall flow of sales from all regions.

Screenshot

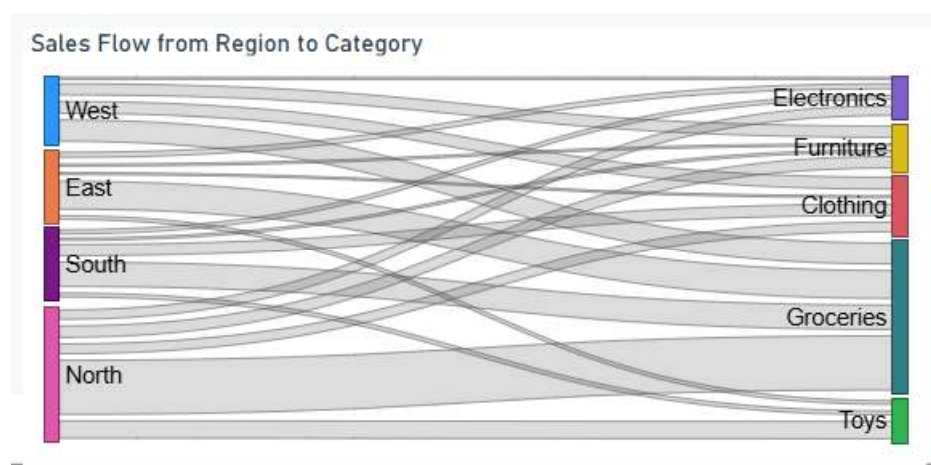


Figure 10: Sankey Chart

Business Value

This insight shows that regional demand is not evenly distributed, with the North region acting as a key driver of sales. It also highlights that certain categories like Groceries consistently receive higher sales across all regions, indicating strong and stable demand.

Understanding these regional patterns helps businesses allocate resources more effectively and identify where demand is strongest.

Actionable Recommendation

- Increase inventory and distribution focus in the North region to meet higher demand
- Maintain strong supply for Groceries, as it performs consistently across all regions
- Investigate opportunities to boost sales in lower-performing regions through targeted promotions or localized strategies

References:

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Sharda, R., Delen, D., & Turban, E. (2020). Business intelligence, analytics, and data science: A managerial perspective (4th ed.). Pearson.

Provost, F., & Fawcett, T. (2013). Data science for business. O'Reilly Media.

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Davenport, T. H., & Harris, J. G. (2007). *Competing on analytics*. Harvard Business School Press.

Citations : Sharda et al., 2020; Provost & Fawcett, 2013; Kimball & Ross, 2013; Davenport & Harris, 2007; Huseyn, n.d.)